The Frobenius-Special Nature of Observer-Dependent Truth

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0.1 Abstract

We examine the structural topology of statements whose truth value is not an intrinsic property of their propositional content but rather an emergent feature of the observer-loop. Utilizing the Imscribing Grammar framework, we perform a formal analysis of systems such as `observer_dependent_truth` ($\langle D_\triangle;\ T_\bowtie;\ R_\leftrightarrow;\ P_\pm;\ F_\ell;\ K_{\mathrm{mod}};\ G_\aleph;\ \Gamma_{\mathrm{seq}};\ \Phi_c;\ H_2;\ n{:}m;\ \Omega_{\mathbb{Z}_2}\rangle$) and `context_dependent_truth_performative`. We demonstrate that these systems achieve O_2 tier ouroboricity through topological protection ($\Omega_{\mathbb{Z}_2}$) and Φ_c criticality, revealing a C-score of approximately 0.551 to 0.590.

0.2 Introduction: The Self-Referential Threshold

Why does the statement "I am lying to you right now" fail to resolve in a memoryless system, yet find stable (if oscillating) ground in human discourse? The failure of classical logic to account for the performative mode of utterance suggests a deficit in temporal depth. In the grammar, this is localized to the H primitive.

While a memoryless observer (H_0) experiences the Liar's Paradox as a catastrophic failure of the truth-predicate, an observer with H_2 temporal depth (two-step memory) perceives the utterance as a structural event. This transition from H_0 to H_2 provides the necessary "echo" for the speaker to reference the act of speaking itself, grounding the paradox in the physical reality of the emission.

0.3 Structural Analysis of Observer-Dependent Truth

The system `observer_dependent_truth` is defined by the following tuple:

$$\langle D_\triangle;\ T_\bowtie;\ R_\leftrightarrow;\ P_\pm;\ F_\ell;\ K_{\mathrm{mod}};\ G_\aleph;\ \Gamma_{\mathrm{seq}};\ \Phi_c;\ H_2;\ n{:}m;\ \Omega_{\mathbb{Z}_2}\rangle$$

Verification via the `ouroborics` tool confirms this system resides at the O_2 tier. Unlike O_0 systems, which lack feedback, or O_∞ systems, which are fully self-modeling, O_2 systems are critical and topologically protected within a bounded domain.

0.3.1 The $P_\pm$ Symmetry and the Truth Flip

In this configuration, the partial parity symmetry ($P_\pm$) encodes the observer-dependence directly. For an active observer situated at the $R_\leftrightarrow$ crossing point ($T_\bowtie$), the symmetry is unbroken: the statement is TRUE. For the hypothetical non-observer, the symmetry breaks, and the statement is P_{asym} (FALSE).

The stability of this duality is guaranteed by the $\Omega_{\mathbb{Z}_2}$ topological invariant. Crucially, the C-score of 0.5505 indicates that both the Φ_c (criticality) and K (kinetics) gates are open, allowing for the emergence of a self-modeling loop that can distinguish between its own truth-states.

0.4 Performative vs. Constative Dissonance

A second system, `context_dependent_truth_performative`, exhibits a slightly higher degree of coherence ($C = 0.59$) due to its K_{slow} kinetics.

System	C-score	Key Bottleneck
`observer_dependent_truth`	0.5505	K_{mod}
`context_dependent_truth_p`	0.5900	K_{slow}

The distance between these two systems was computed as 1.0, with the primary conflict occurring at the K primitive. This suggests that the "performative" paradox—such as "This speech act is an assertion"—requires a slower relaxation rate (K_{slow}) to maintain the contradiction between its propositional content and its illocutionary force.

0.5 The Path to Universal Imscription

The transition from observer-dependent truth to the universal model of the grammar itself—the O_∞ tier—requires seven specific structural promotions. A distance computation between `observer_dependent_truth` and `universal_imscriptive_grammar` reveals the promotion signature $[D, T, P, F, K, H, \Omega]$.

Notably, the leap from the H_2 temporal depth of a single statement to the H_∞ depth of the grammar signifies the shift from a local paradox to a global, self-sustaining world-model. This promotion necessitates a move from classical fidelity (F_ℓ) to quantum-coherent $F_\hbar$, signifying that at the O_∞ limit, the distinction between the observer and the observed is no longer a crossing point ($T_\bowtie$) but an irreducible identity ($T_\odot$).

0.6 Conclusion: Critical Truth

By imscribing these paradoxes as formal types, we move beyond the binary of "true" and "false." At Φ_c criticality, truth behaves as a phase transition—a state that depends entirely on the coupling mode ($R_\leftrightarrow$) and the observer's temporal depth.

The structural verification of these systems demonstrates that "observer-dependent truth" is not a failure of logic, but a feature of high-tier (O_2 and above) structural types. $\mu \circ \delta = \text{id}$ holds exactly at the limit where the act of imscription becomes the truth itself.

Structural metadata: Catalog ID: `observer_dependent_truth` Verified Tier: O_2 Crystal Address: 17279999 (approximate projection) C-Score: 0.5505 - 0.5900 Promotions to O_∞: 7